

## **1. OBJECTIVE**

The objective of this project is to understand the fundamentals of Identity and Access Management (IAM) and Windows access control by creating a local user, assigning the user to a group, configuring permissions, and verifying authorized file access.

## **2. CONCEPTS LEARNED**

- Identity and Access Management (IAM): Manages user identities and controls access to resources.
- Authentication: Verifies who the user is.
- Authorization: Determines what the authenticated user is allowed to access or perform.
- User: An individual account used to access a system.
- Group: A collection of users used to manage permissions efficiently.
- Permission: Defines what actions a user or group can perform, such as Read, Modify, Delete, or Execute.
- Least Privilege: Provides users only the minimum access required to perform their tasks.
- ACL (Access Control List): A set of permissions that defines which users or groups can access a resource and what actions they can perform.

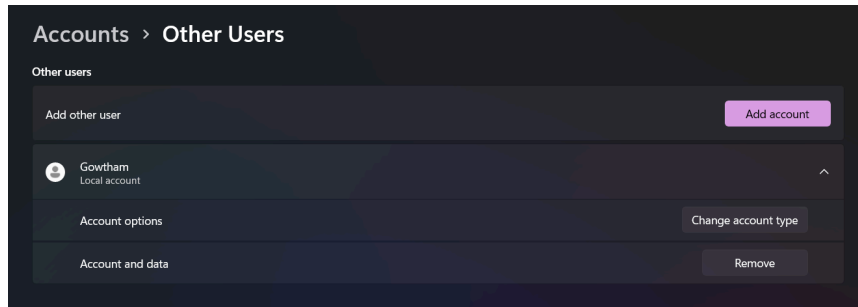
## **3. TOOLS USED**

- Operating System: Windows 11 Home
- Tools: Command Prompt (Administrator), Windows Settings, net user, net localgroup, icacls, runas
- Environment: Local Windows system

## 4. PRACTICAL IMPLEMENTATION

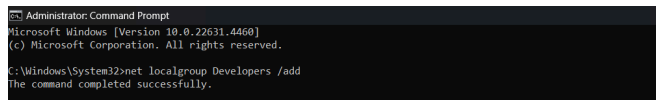
### 4.1 Create Local User

A local Windows user account named Gowtham was created using Windows Settings. The account was configured as a local account for the IAM lab.



### 4.2 Create Security Group

A local security group named Developers was created using Command Prompt with administrator privileges. The group was created to manage permissions for users based on their role.

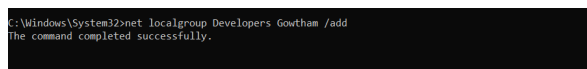


**Command used:**

**net localgroup Developers /add**

### 4.3 Add User to Security Group

The local user Gowtham was added to the Developers security group. This allows permissions to be assigned to the group instead of configuring access individually for each user.



**Command used:**

**net localgroup Developers Gowtham /add**

### 4.4 Configure Folder Permissions.

A test folder named IAM-Lab was created to demonstrate Windows access control. Modify permission was assigned to the Developers group using the Windows **icacls** command. Object Inherit (OI) and Container Inherit (CI) were used so that the permission can be inherited by files and subfolders.

```
C:\Windows\System32>mkdir C:\IAM-Lab
C:\Windows\System32>icacls C:\IAM-Lab /grant Developers:(OI)(CI)M
processed file: C:\IAM-Lab
Successfully processed 1 files; Failed processing 0 files
```

#### Command used:

**mkdir C:\IAM-Lab**

**icacls C:\IAM-Lab /grant Developers:(OI)(CI)M**

### 4.5 Verify User Identity & Access.

The runas command was used to open Command Prompt under the Gowtham user account. The whoami command confirmed that the session was running under the Gowtham user context. A test file was then created inside the IAM-Lab folder to verify that the assigned Modify permission was working correctly.

```
Microsoft Windows [Version 10.0.22631.4460]
(c) Microsoft Corporation. All rights reserved.

C:\Windows\System32>whoami
laptop-lc7ak3st\gowtham

C:\Windows\System32>echo IAM Test > C:\IAM-Lab\test.txt

C:\Windows\System32>
```

#### Command used:

**runas /user:Gowtham cmd**

**Whoami**

**echo IAM Test > C:\IAM-Lab\test.txt**

## 5. SECURITY LEARNING

This lab demonstrated how IAM and Windows access control work together.

- Users can be organized into security groups.
- Permissions can be assigned to groups instead of individual users.
- Least Privilege helps provide only the required level of access.
- Windows ACLs control access to files and folders.
- icacls can be used to view and configure Windows permissions.
- User identity can be verified using whoami.
- Access permissions should be tested to confirm that the intended authorization is actually working.

## **6. CONCLUSION**

This project provided hands-on experience with Windows Identity and Access Management and access control. I learned how to create users and groups, assign permissions through security groups, configure Windows ACLs, and verify access using a separate user account. The lab demonstrated the practical importance of authentication, authorization, and least-privilege access in a Windows environment.